HAPPIEST HEALTH PREVENTIVE ENDOCRINOLOGY CLINIC

BENGALURU, INDIA

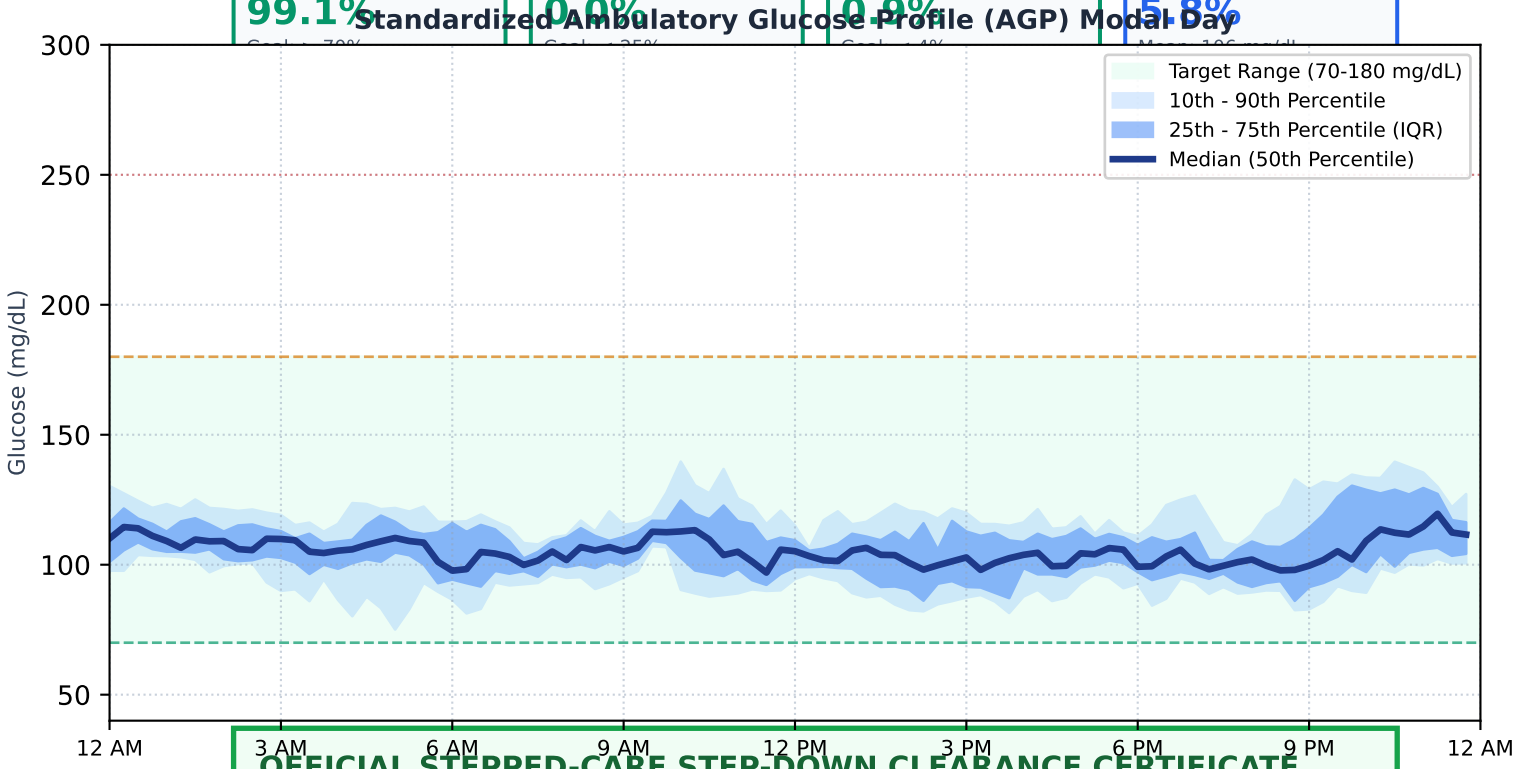
Ambulatory Glucose Profile (AGP) & Stepped-Care Transition Certificate

Patient ID: TT-IND-001 | ABHA ID: 91-9999-0001-0000

Age: 27 yrs | Sex: M | Phenotype: Normoglycemic | Baseline Lab HbA1c: 5.4%

Induction: 7-Day CGM Completed

TIME IN RANGE (70-180)	TIME ABOVE RANGE (>180)	TIME BELOW RANGE (<70)	ESTIMATED GMI (HbA1c)
99.1%	**0.0%**	**0.9%**	**5.8%**



OFFICIAL STEPPED-CARE STEP-DOWN CLEARANCE CERTIFICATE

PROTOCOL: Calibrate-Then-Step-Down (7 Days CGM -> Tier 2 SMBG Maintenance)

1. CLINICAL STABILITY: Patient TT-IND-001 has successfully accumulated 7 days of blinded CGM continuous profile. Observed Time in Range is 99.1% with CV of 12.9%.
2. DIGITAL TWIN READY: The patient's metabolic twin has achieved a shrinkage weight of w = 0.90, capturing personalized carbohydrate sensitivity without requiring continued INR 10,000/month sensor expenditure.
3. STEP-DOWN APPROVED: The patient is clinically cleared to transition to Tier 2 Frugal Glucometer Care. Prescribed Schedule: 3-4 capillary fingersticks/day (Fasting, 2h Post-Breakfast, 2h Post-Dinner).
4. ECONOMIC IMPACT: Direct consumable savings of INR 8,800/month (INR 1,05,600 / year) retained by the household.

Dr. V. Lomate, MD (Endocrinology)
Consulting Diabetologist, Happiest Health

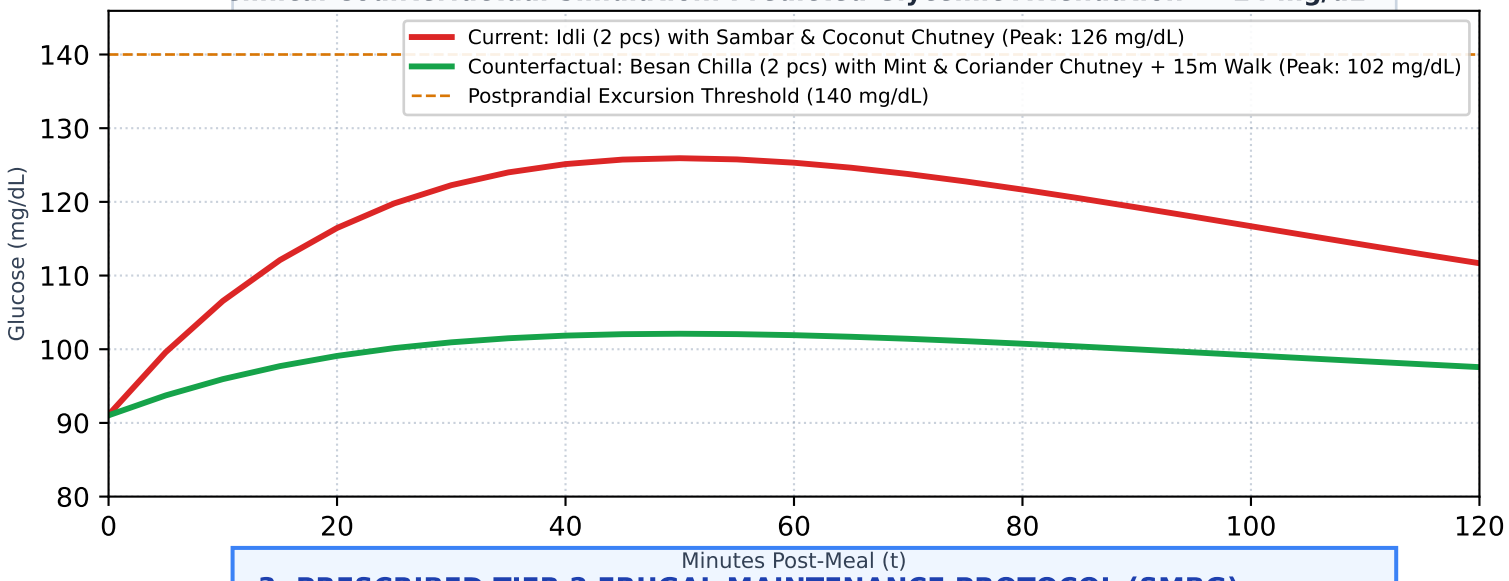
THALITWIN DIGITAL TWIN PERSONALIZATION & NUTRITION PRESCRIPTION

Empirical-Bayes Metabolic Sensitivity & Indian Food What-If Counterfactuals

1. DIGITAL TWIN METABOLIC CALIBRATION AUDIT

- Calibration Period: 7 Days | Observed Valid Meals: 36 | Prior Model: LightGBM + Population EHR
- Shrinkage Weight (w): 0.900 (Formulation: w = n / (n + 4.0), blending individual vs population prior)
- Calibrated Carb Sensitivity: 0.33 mg/dL per gram carbohydrate (Baseline Population Prior: 0.80 mg/dL/g)
- Personal Residual Glycemic Offset: -9.7 mg/dL
- Conformal Uncertainty Ribbon: Split-conformal 90% confidence band calibrated at ±55.5 mg/dL.

Clinical Counterfactual Simulation: Predicted Glycemic Attenuation = -24 mg/dL



3. PRESCRIBED TIER 2 FRUGAL MAINTENANCE PROTOCOL (SMBG)

- MORNING FASTING CHECK (07:30 AM): Baseline capillary reading (Goal: 80 - 110 mg/dL).
- POST-BREAKFAST CHECK (2 Hours Post-Meal): Evaluates breakfast spike attenuation (Goal: < 140 mg/dL).
- POST-DINNER CHECK (2 Hours Post-Meal): Confirms glycemic recovery before nocturnal rest.
- ACTIVE INQUIRY SAFETY NET: If the patient consumes an unlogged snack, ThaliTwin's accelerometer and glucose drift monitor automatically alerts: 'Did you just have tea or a snack?' to re-center the twin.
- EMERGENCY CLINICAL DRIFT TRIGGER: If fasting glucose > 180 mg/dL for 3 consecutive days, the twin initiates an automated tele-triage alert to the Happiest Health care coordinator.

4. DIGITAL HEALTH PROVENANCE & REGULATORY COMPLIANCE

- ABDM Gateway Integration: Bundle ID abdm-bundle-pat-001 · NRCES NDHM FHIR R4 standard.
- Clinical SNOMED-CT Coding: 73761001 (Stepped-Care Type 2 Diabetes Glycemic Care Plan).
- Observation Provenance: ISO-15197 compliant glucometer test strips · LOINC 15074-8 Interstitial Glucose.
- DPDP Act (India 2023): Clinical Decision Support prototype. No automated prescription alterations.

Validated by Clinician:
Happiest Health Digital Twin Team
Bengaluru Summit Track 2026

CONFIDENTIAL MEDICAL RECORD